Sprint 2 Data Flow Specification

Context diagram, level one flows, data contracts, and trust boundaries

IRON CHEF v3 | Vo Trung Tin | Reviewed 11 October 2026

This document traces data among the browser, frontend adapters, monolith, and Lead-owned AI or evidence sources. Each flow identifies inputs, transformations, stores, and failure states. A diagram arrow expresses an interface or requirement, not proof that an external model is accepted.

Source baseline and interpretation

Source	Evidence used
Sprint 2 report supplied by the user	Work assignment: Vo Trung Tin owns UI requirements, use cases, data flows, and Member scaffold acceptance. The sprint runs 23 September–10 October 2026.
Member backlog, branch codex/member-sprint12-deliverables	Epic 1–12 implementation states and open gates G1–G10; engineering checkpoint dated 6 October 2026.
Lead backlog v109, origin/master 8bb8167	As of 9 October 2026, Stage 3 is artifact-verified through exact step 327/561. Segment 6 was launched toward step 374. Final model quality and scoring remain unverified.
Browser captures dated 6 October 2026	Actual frontend views of /explore/graph and /scan, with the graph backed by local seed data.
Frontend and monolith code	Routes, request boundaries, error states, fields, and adapter behavior reviewed for this edition.

Evidence rule: a browser image proves that a screen appeared during that run; code shows an implemented path; a backlog records the team's checkpoint. None alone proves production deployment, model quality, clinical safety, or a research result.

1 Notation and trust boundaries

Term	Meaning
External entity	User, Creator, Reviewer, or Lead source that supplies or receives data.
Process	UI or service step that validates and transforms input.
Store	Profile, recipe, graph, receipt/event, or evidence manifest data.
Trust boundary	Browser → same-origin frontend → monolith or model provider. User bearer credentials are not forwarded directly to a model provider.

2 Context diagram

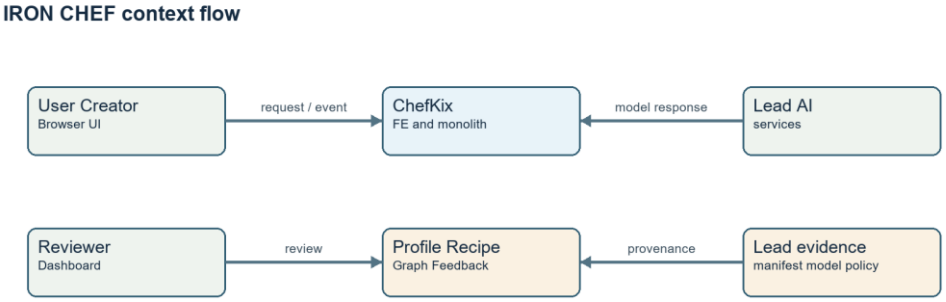


Figure 1. Logical context; detailed request and response paths are specified below.

3 Level one processes

Process	Inputs and outputs
P1 Recipe and profile	Creator recipe and User allergen settings → stored recipe, normalized profile, and context for P3.
P2 Photo detection	Camera/upload image plus caller identity → sourced detections or explicit integration-pending state.
P3 Substitution and safety	Ingredient, recipe context, tags, flags, candidate policy → three-valued card and real candidate receipt.
P4 Graph and compounds	Search/root/depth/limit or ingredient pair → bounded graph with coverage and grounded evidence where available.
P5 Feedback	Session, candidate receipt, choice/outcome → durable event, projection, ingestion, and retry identity.
P6 Evaluation and thesis	Lead manifest and artifact references → protected dashboard, figure caption, and evidence readiness state.

4 DF 01 Photo to ingredients and recipe candidates

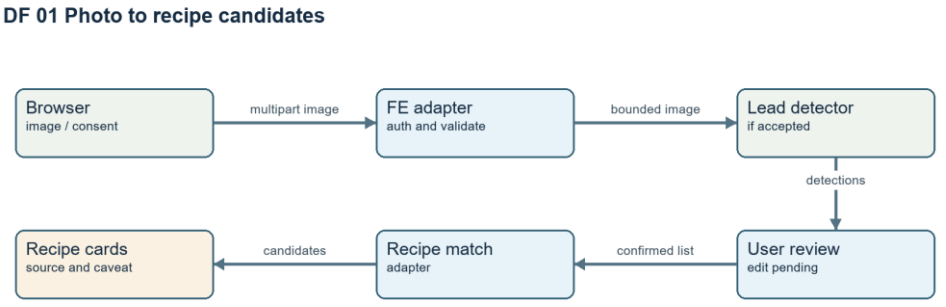


Figure 2. User correction is an open UI gate and must occur before downstream action in the accepted flow.

Step	Input, validation, and failure behavior
Browser	Request camera only after action or accept upload. Denied or unsupported camera offers upload or manual input; a guest screenshot does not prove a device-camera run.
Frontend adapter	Authenticate caller; validate multipart image type, non-zero bytes, and ≤10 MiB. Propagate cancellation and cap upstream work at 30 seconds. Return 400/413 or 503 as appropriate.
Lead detector	Return name, box, confidence, and source only from an accepted configured model. M3 is not yet accepted; malformed or failed response is an error, never fabricated output.
User review	Correct or remove detections and confirm the list. The observed /scan screen does not yet establish this editing control.
Recipe matching	Submit no more than 100 ingredient names of up to 120 characters each through the protected bounded route. Accepted M4 model/index and rights evidence remain open.

5 DF 02 Profile to substitution and safety

Authenticated settings → monolith profile store → normalized allergen flags → substitution request with ingredient, reason, recipe_context, dietary_tags, flags, and session_id → candidate/policy/receipt response → frontend resolver → Safe, Check, or Blocked card → selection or decline. Missing or unreviewed policy never becomes Safe. Custom or unknown terms remain Check without adjudication.

Data	Source and handling
Allergen flags	Saved under the authenticated profile, normalized, forwarded in later requests, and excluded from public evidence logs.
Policy result	Reviewed policy source only. Missing or malformed → Check; conflict → Blocked; Safe only with sufficient reviewed evidence.
Candidate receipt	Preserve source identity for feedback; never synthesize a missing receipt.
Compounds and nutrition	Separate official evidence path with source, grounding, and units. Null means unavailable and never upgrades allergen safety.

6 DF 03 Knowledge graph and compounds

Browser search or root → GET /knowledge/graph with depth 0–2 and limit 1–500 → seed or imported monolith collection → bounded nodes, edges, and coverage → client focus, pan, zoom, and detail. The captured Ginger result has no edge. The four seed edges are substitutions only. Filter labels for chemical similarity and co-occurrence do not create scientific data. Compound profile and pair requests use a separate official source path; ratio is not confidence and overlap does not prove culinary suitability.

7 DF 04 Feedback and research evidence

Cooking session → choice/outcome event with candidate receipt → monolith durable write → projection and AI ingestion/replay. Failed writes, retries, and replay retain event identity and cannot be reported as model learning. Separately, Lead dataset/split/protocol/prediction artifacts → manifest and review decision → frontend schema validation → dashboard/export caption → thesis artifact. The frontend validates structure, not the independent identity, rights, review, or correctness of upstream science.

Store	Owner and current state
Profiles and recipes	Monolith product storage and frontend flows exist; deployed authenticated acceptance remains to be checked.
Knowledge graph	Monolith currently serves a 25-ingredient, four-edge seed. Official G5 import is pending.
Receipts and feedback events	AI and monolith paths exist; representative cross-service outcome and recovery evidence remains pending.
Evaluation manifests	Lead-owned. Frontend schema and captions exist; verified artifacts and scored results remain pending.
Thesis captures	Member and Lead reviewers. Fourteen chapters are mapped; individual source-backed captures and reviews remain pending.

8 Data safety and closure matrix

Flow	Evidence required before closure
Photo	Real devices, provider authentication, accepted detector and retrieval model/index manifests, and image rights. Screen controls are not inference evidence.
Safety	Reviewed policy, authenticated deployed profile-to-policy-to-UI test, and adjudicated benchmark. Compound overlap and absence of a match do not establish Safe.
Graph	Rights/schema/hash-validated import, coverage checks, rollback, and API-backed captures. A seed API is not the canonical graph.
Evaluation	Actual Lead artifacts and checked hashes, split/protocol/seeds, intervals, abstention, and review. A training checkpoint is not a quality score.

Lead v109 verifies secondary Stage 3 through exact step 327/561 and records a Segment 6 launch toward step 374. Completion of Stage 3, inference, scoring, and model quality remain open.